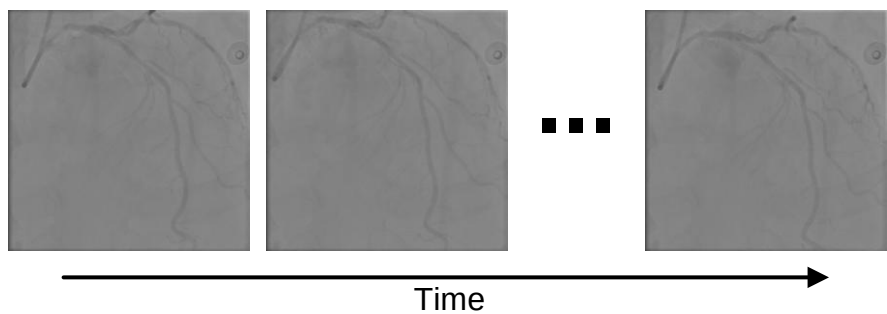
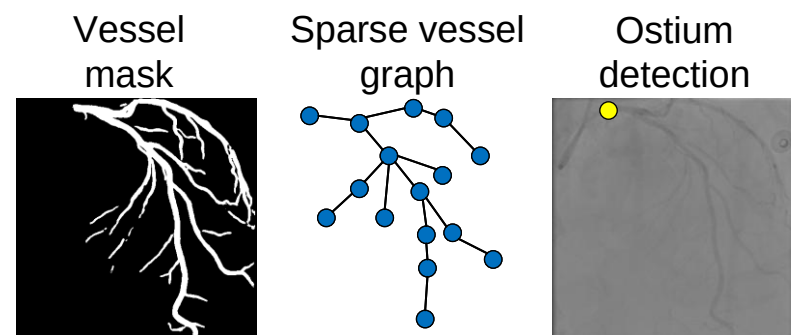


A Offline / Online Preprocessing

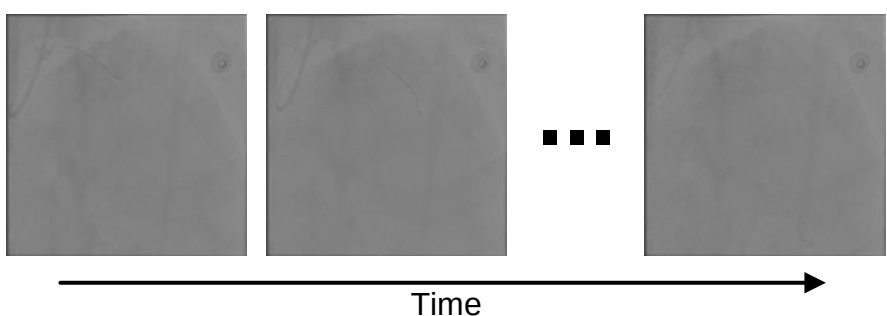
Offline angiography sequence



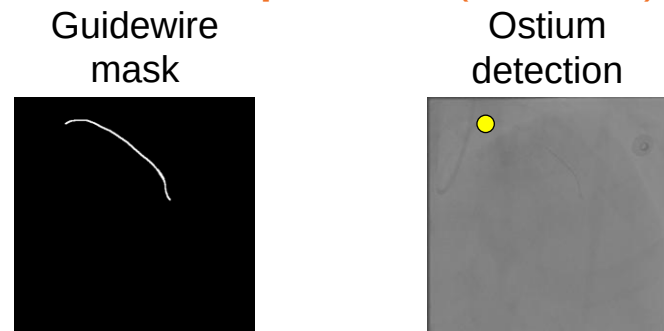
Offline products (extracted)



Online guidewire sequence

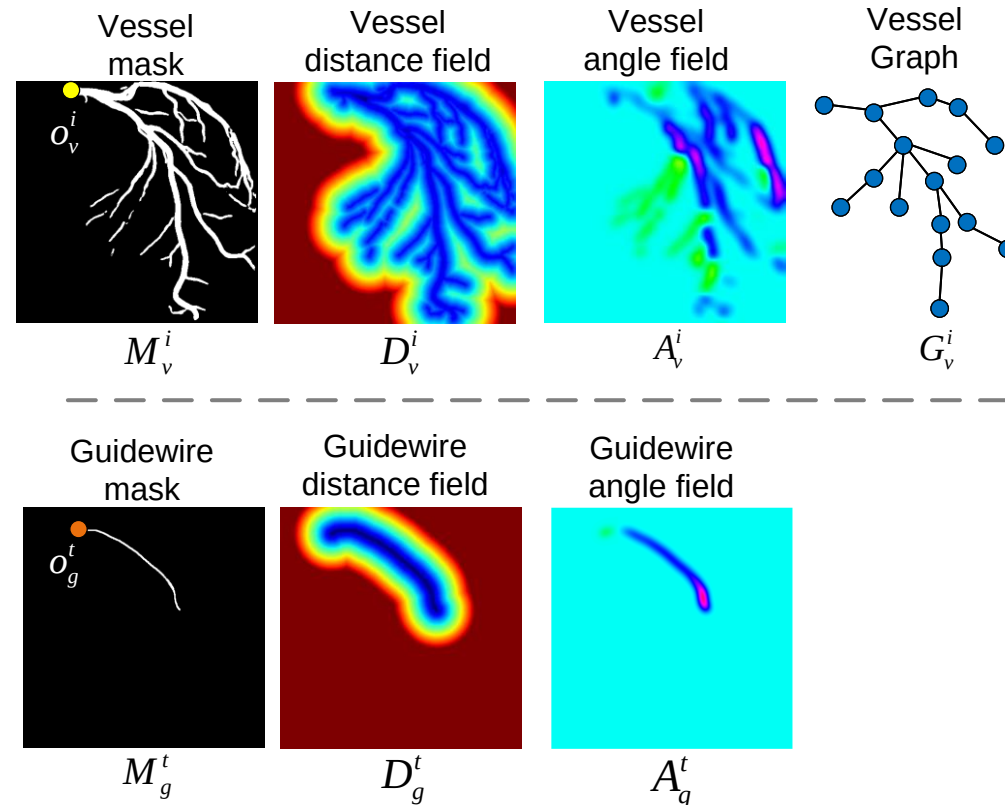


Online products (extracted)



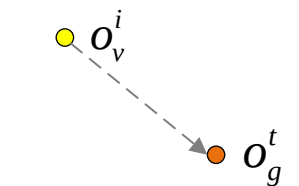
B Ostium-based Coarse Alignment

Before alignment



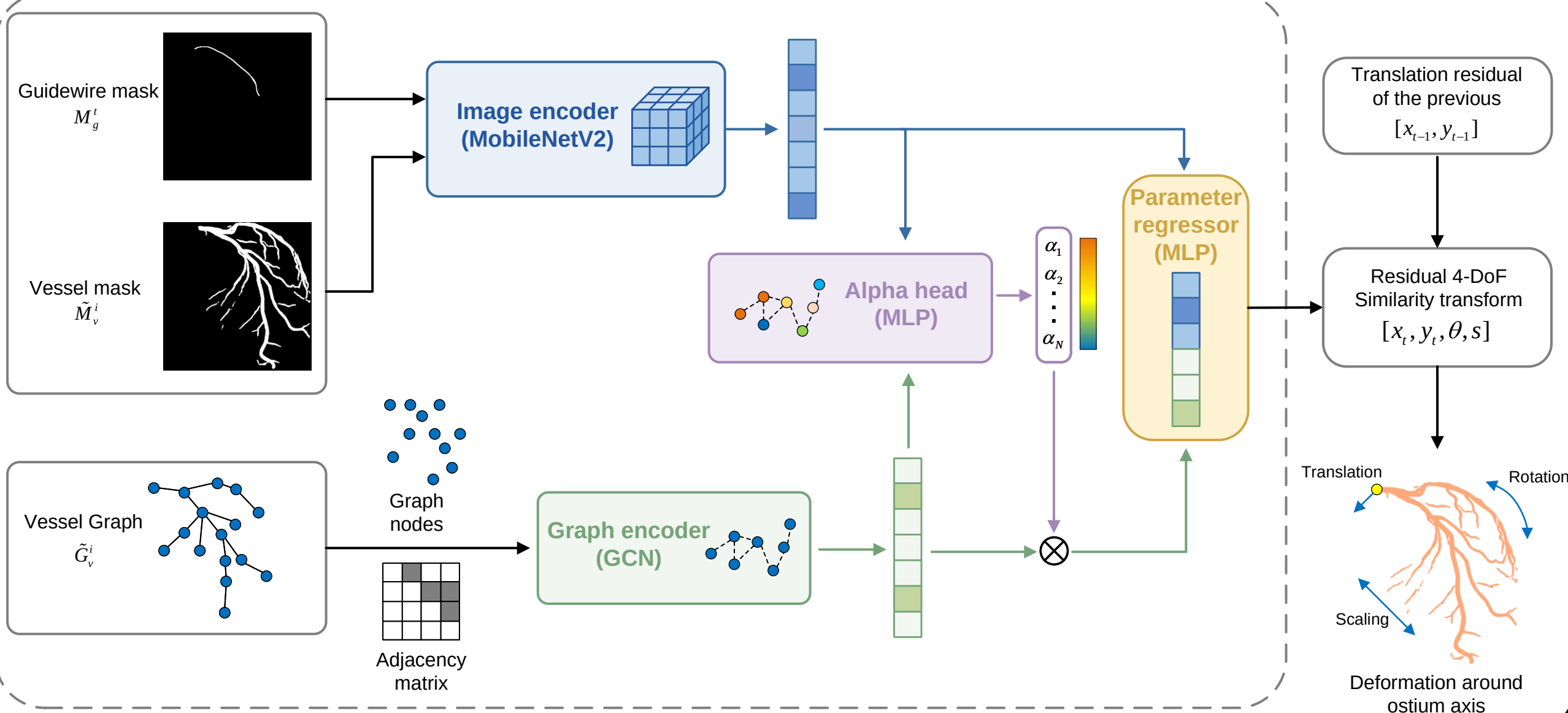
Ostium translation

$$d_{i,t} = o_g^t - o_v^i$$



o_v^i : vessel ostium
 o_g^t : guidewire ostium

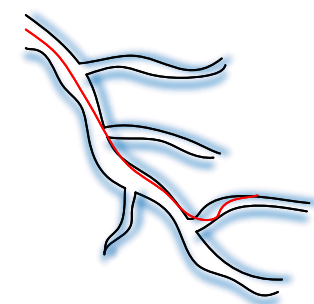
C Graph-guided Similarity Registration Network



D Training Objectives

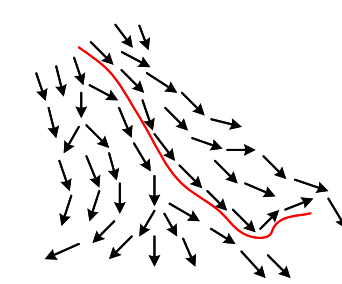
1 **Containment loss**
(keep guidewire inside lumen)

Signed distance field (derived)



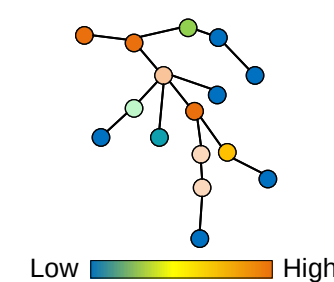
2 **Tangential consistency loss**
(align local direction)

Tangential angle field (derived)

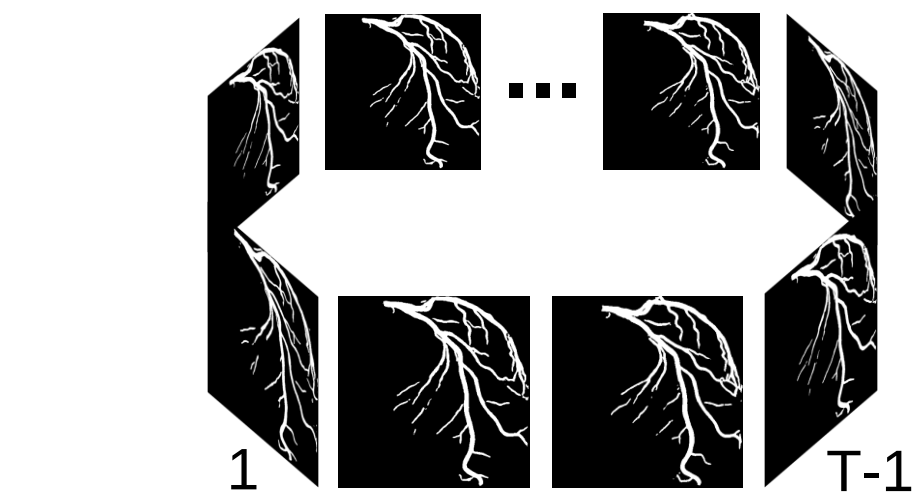


3 **Graph smoothness / node consistency loss**

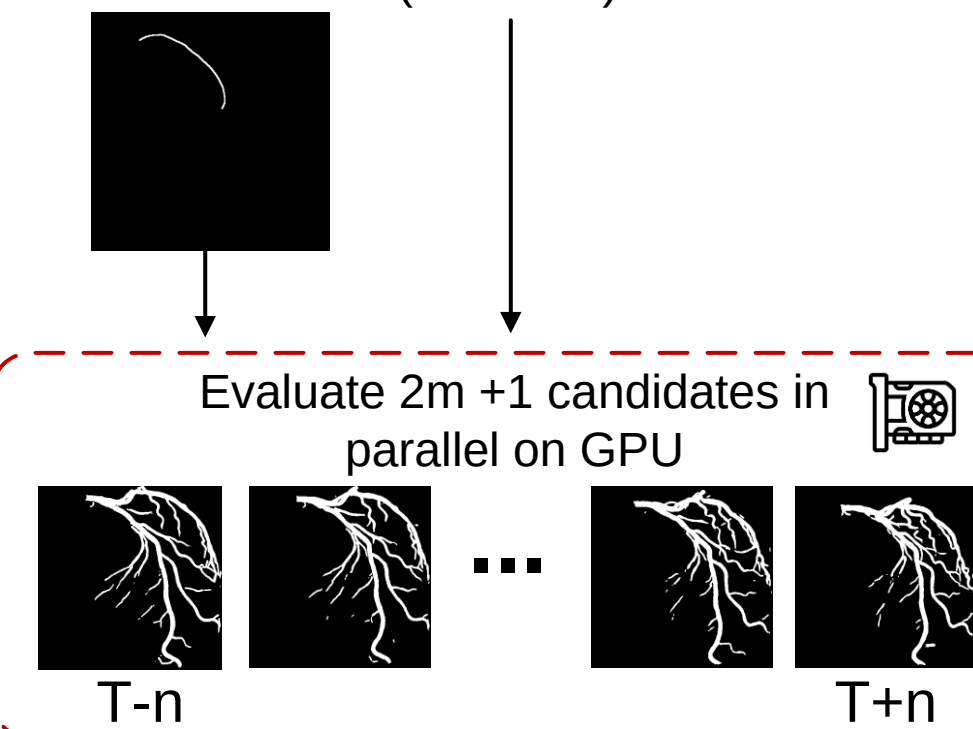
Higher weights near guidewire with neighborhood smoothness (derived)



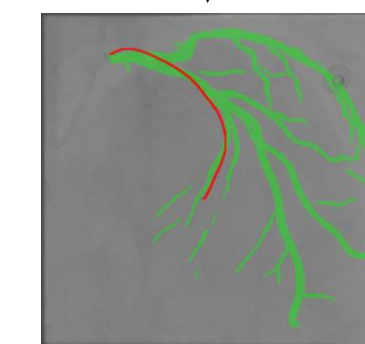
E ECG-free Online Roadmap Selection and Output



Evaluate 2m + 1 candidates in parallel on GPU



Scoring (similarity score)



Best candidate